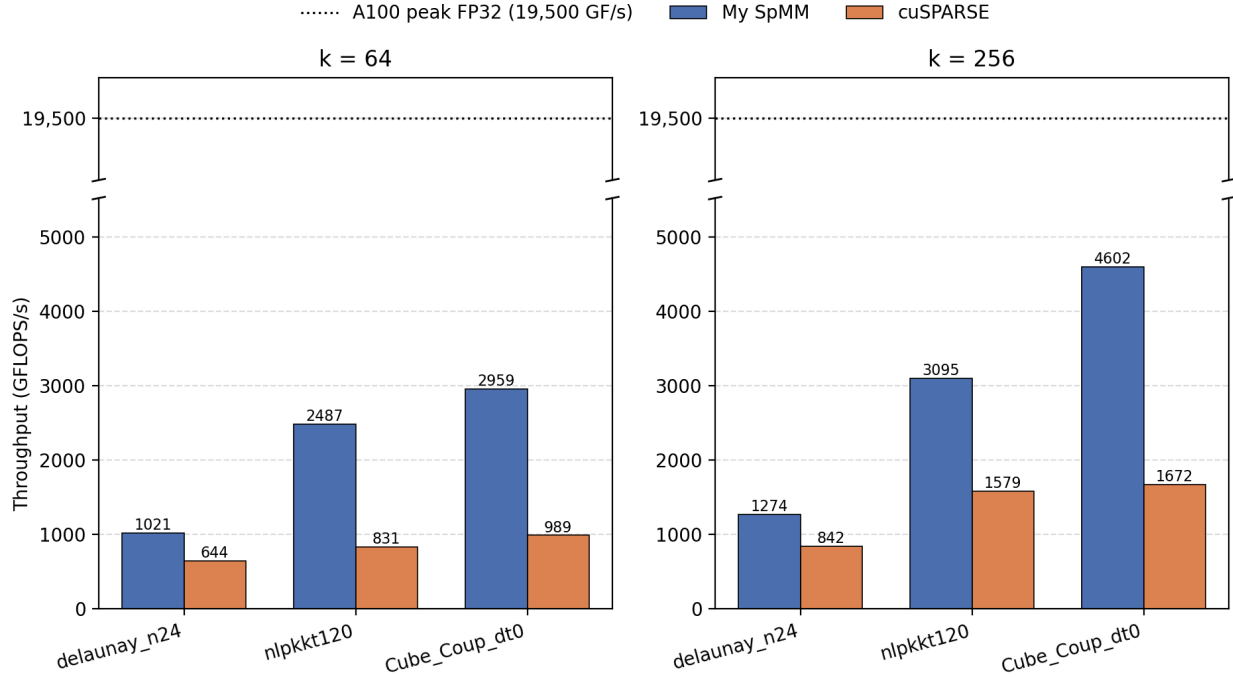


SpMM throughput vs cuSPARSE on NVIDIA A100 (40 GB)



SpMM throughput of our warp-per-row kernel vs NVIDIA cuSPARSE's `cusparsespMM` on three SuiteSparse matrices, measured on a single 40 GB A100 at $k=64$ and $k=256$. The dotted line marks A100 peak FP32 throughput (19,500 GFLOPS/s). Our kernel outperforms cuSPARSE on all three matrices, with speedups of roughly 1.5–3 $\times$, and improves further at $k=256$ because each sparse load of A is amortized over 4 $\times$ more arithmetic. Performance varies across matrices likely due to differences in sparsity structure: `del aunay_n24` has scattered, near-random nonzero locations with little reuse of X , giving the lowest throughput; `Cube_Coup_dt0` has block-banded structure that yields high cache reuse and the best throughput; `nlpkkt120` falls in between. All results sit well below the peak compute ceiling.